

Vivek Notes For Mounting EFS with Two different EC2 and Different Availability Zone

Step 1 — Create two EC2 instances

note :- all commands highlighted with green color

Create:

EC2-1

- Ubuntu
- AZ: ap-south-1a
- Security Group : Choose NFS

EC2-2

- Ubuntu
- Availability Zone : ap-south-1b :- **the availability zone choose while making subnet and same instance choose for that availability zone**
- Security Group : Choose NFS

They can be in the **same VPC**.

Important

Both EC2 instances need network connectivity to EFS.

For this practical, using the **same VPC** makes things easier.

Step 2 — Create EFS

Go to:

AWS Console → EFS → File systems → Create file system

Click:

Create file system

Select your VPC.

For example:

VPC: Vivek-VPC

Then create the file system.

Step 3 — Connect to EC2-1

- After connect EC2 go to root user using **sudo su -**
- Then follow that steps or commands to install NFS in that serve/instance
- Command one - 1) **sudo apt update**
- Command two - 2) **sudo apt install nfs-kernel-server -y**
- Command three - 3) **sudo systemctl status nfs-kernel-server** to check status of NFS is Enable or not
- Command four - 4) **sudo systemctl start nfs-kernel-server** if not enable run this command
- Then make directory – 5) **mkdir efs**

- Run the command is shown in the EFS section we try to attach it and also remember one thing select the **attach via IP** select this point
- For example that command is :- `sudo mount -t nfs4 -o nfsvers=4.1,rsize=1048576,write=1048576,hard,timeo=600,retrains=2,noresvport 192.178.10.244:/ efs`
- Go to that directory and make file for example :- `touch vivek.txt`
- To check :- `ls`

Same process for second instance

Step 4 — Connect to EC2-2

- After connect EC2 go to root user using `sudo su -`
- Then follow that steps or commands to install NFS in that server/instance
- Command one - 1) `sudo apt update`
- Command two - 2) `sudo apt install nfs-kernel-server -y`
- Command three - 3) `sudo systemctl status nfs-kernel-server` to check status of NFS is Enable or not
- Command four - 4) `sudo systemctl start nfs-kernel-server` if not enable run this command
- Then make directory - 5) `mkdir efs`
- Run the command is shown in the EFS section we try to attach it and also remember one thing select the **attach via IP** select this point
- For example that command is :- `sudo mount -t nfs4 -o nfsvers=4.1,rsize=1048576,write=1048576,hard,timeo=600,retrains=2,noresvport 192.178.10.244:/ efs`
- Go to directory : `cd efs`
- To check :- `ls`

To connect EFS steps

Step 1 :- click on efs that you created

Step 2 :- scroll down and click on network

Step 3 :- click on manage

Step 4 :- find your availability zone and choose your instance wizard type

The wizard type is available in your EC2 instance click on security and scroll down

Step 5 :- click on attach

Step 6 :- select Mount Via IP

Step 7 :- choose availability zone

Step 8 :- copy that command which you see in section (using the NFS client)

Step 9 :- paste that to mount with NFS server

- First EC2 instance ScreenShot for reference after mounting

```
Scanning linux images...
Running kernel seems to be up-to-date.
No services need to be restarted.
No containers need to be restarted.
No user sessions are running outdated binaries.
No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@ip-192-178-10-22:~# sudo systemctl status nfs-kernel-server
● nfs-server.service - NFS server and services
   Loaded: loaded (/usr/lib/systemd/system/nfs-server.service; enabled; preset: enabled)
   Active: active (exited) since Thu 2026-09-24 12:56:05 UTC; 19s ago
  Invocation: 70f4cc7497924e1fb7d14b476d4eae58
     Docs: man:rpc.nfsd(8),
           man:exportfs(8),
           Main PID: 2557 (code=exited, status=0/SUCCESS)
    Mem peak: 2M
       CPU: 20ms

Sep 24 12:56:05 ip-192-178-10-22 systemd[1]: Starting nfs-server.service - NFS server and services...
Sep 24 12:56:05 ip-192-178-10-22 exportfs[2555]: exportfs: can't open /etc/exports for reading
Sep 24 12:56:05 ip-192-178-10-22 systemd[1]: Finished nfs-server.service - NFS server and services.
root@ip-192-178-10-22:~# mkdir /efs
root@ip-192-178-10-22:~# sudo mount -t nfs4 -o nfsvers=4.1,rsize=1048576,wsize=1048576,hard,timeo=600,retrans=2,noresvport 192.178.10.244:/ efs
mount.nfs4: mount point efs does not exist
root@ip-192-178-10-22:~# ls
snap
root@ip-192-178-10-22:~# mkdir efs
root@ip-192-178-10-22:~# sudo mount -t nfs4 -o nfsvers=4.1,rsize=1048576,wsize=1048576,hard,timeo=600,retrans=2,noresvport 192.178.10.244:/ efs
root@ip-192-178-10-22:~# cd efs
root@ip-192-178-10-22:~/efs# touch vivek.txt
root@ip-192-178-10-22:~/efs# ls
vivek.txt
root@ip-192-178-10-22:~/efs# cd ..
root@ip-192-178-10-22:~#
```

i-07ec2ba2df1819e7b (EC2-1)
PublicIPs: 52.53.125.24 PrivateIPs: 192.178.10.22

- Second EC2 instance ScreenShot after mounting

```
nfs-mountd.service is a disabled or a static unit, not starting it.
nfsdclld.service is a disabled or a static unit, not starting it.
Creating config file /etc/exports with new version
Creating config file /etc/default/nfs-kernel-server with new version
Processing triggers for man-db (2.13.1-1build1) ...
Processing triggers for libc-bin (2.43-2ubuntu2) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@ip-192-178-20-169:~# sudo systemctl status nfs-kernel-server
● nfs-server.service - NFS server and services
   Loaded: loaded (/usr/lib/systemd/system/nfs-server.service; enabled; preset: enabled)
   Active: active (exited) since Thu 2026-09-24 13:06:43 UTC; 16s ago
  Invocation: 3cd4f5cee8bb4acc8f2dc48ef0182165
     Docs: man:rpc.nfsd(8),
           man:exportfs(8),
           Main PID: 2560 (code=exited, status=0/SUCCESS)
    Mem peak: 2M
       CPU: 17ms

Sep 24 13:06:43 ip-192-178-20-169 systemd[1]: Starting nfs-server.service - NFS server and services...
Sep 24 13:06:43 ip-192-178-20-169 exportfs[2558]: exportfs: can't open /etc/exports for reading
Sep 24 13:06:43 ip-192-178-20-169 systemd[1]: Finished nfs-server.service - NFS server and services.
root@ip-192-178-20-169:~# mkdir vivek
root@ip-192-178-20-169:~# sudo mount -t nfs4 -o nfsvers=4.1,rsize=1048576,wsize=1048576,hard,timeo=600,retrans=2,noresvport 192.178.10.244:/ vivek
root@ip-192-178-20-169:~# cd vivek
root@ip-192-178-20-169:~/vivek# ls
vivek.txt
root@ip-192-178-20-169:~/vivek#
```

i-098496692539ca1de (EC2-2)
PublicIPs: 54.176.80.53 PrivateIPs: 192.178.20.169

VPC | us-west-1

Instances | EC2 | us-west-1

EFs | us-west-1

NFS installation guide

us-west-1.console.aws.amazon.com/efs/home?region=us-west-1#/file-systems/fs-008e8af2a32378f89?tabId=mounts

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File systems

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AWS DataSync

AWS Transfer

Documentation

Throughput mode

Elastic

Lifecycle management

Transition into Infrequent Access (IA): 30 day(s) since last access

Transition into Archive: 90 day(s) since last access

Transition into Standard: None

Availability zone

Regional

DNS name

fs-008e8af2a32378f89.efs.us-west-1.amazonaws.com

Replication overwrite protection

Enabled

Metered size

Monitoring

Tags

File system policy

Access points

Network

Replication

Network

Manage

Availability zone (AZ-ID)	Mount target ID	Subnet ID	VPC ID	Mount target state	IPv4 address	IPv6 address	Network interface ID	Security groups
us-west-1a (usw1-az1)	fsmt-04a6a92afb018d6e6	subnet-05a6de5e23627f553	vpc-0c24c08b48d665943	Available	192.178.10.244	N/A	eni-096feaaa928a82070	sg-03b88cf0ce48eb330 (launch-wizard-4)
us-west-1c (usw1-az3)	fsmt-0373f2a26777b7703	subnet-0708789b7f21971b1	vpc-0c24c08b48d665943	Available	192.178.20.156	N/A	eni-087725d1e8b69a53b	sg-0081d59e080472ce6 (launch-wizard-5)

CloudShell

Agent Toolkit for AWS

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Console Mobile App

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Amazon EFS - File system confi...

NFS installation guide

us-west-1.console.aws.amazon.com/efs/home?region=us-west-1#/file-systems/fs-008e8af2a32378f89?tabId=mounts

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Elastic File System

File systems

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AWS DataSync

AWS Transfer

Documentation

Attach

Mount your Amazon EFS file system on a Linux instance. [Learn more](#)

Mount via DNS

Mount via IP

Availability zone

us-west-1a

Using the NFS client:

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sudo mount -t nfs4 -o nfsvers=4.1,rsize=1048576,wsiz=1048576,hard,timeo=600,retrans=2,noresvport 192.178.10.244:/ efs

sudo mount -t nfs4 -o nfsvers=4.1,rsize=1048576,wsiz=1048576,hard,timeo=600,retrans=2,noresvport [MOUNT_TARGET_IPv6]:/ efs

See our user guide for more information. [Learn more](#)

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Monitoring

Tags

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Access points

Network

Replication

Network

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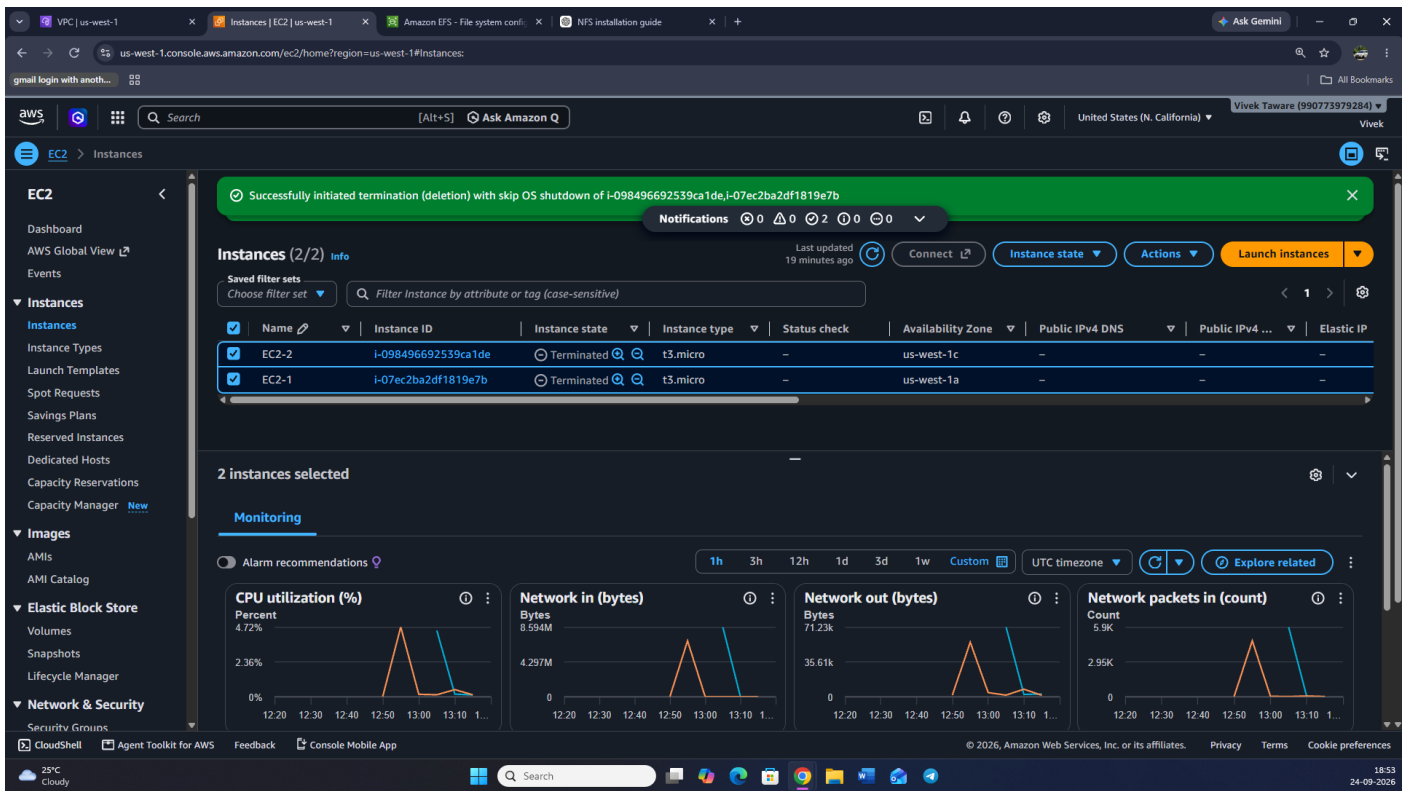
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Use case diagram

